Reference Event 2: The Loss of Power at Memorial Medical Center, August and September 2005

Characterization Against the Healthcare and Public Health Sector Design Basis

DBA-HPH-RE-2 — Reference Event Instrument, DBA-HPH Series

Lineage: Issued under the Healthcare and Public Health Sector Framework v0.6; DBA-MA Parent Framework; conferred on the proposal of Care Delivery (DBA-HPH-CD-FC1)

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Timothy E. Roxey

Eclectic Technologies

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose, Scope and Position

1.1 Purpose

This instrument reads the loss of power at Memorial Medical Center in New Orleans during Hurricane Katrina, from 29 August to 1 September 2005, back through DBA-HPH-CD-FC1 (*Care Delivery*), whose category of the loss of electric power beyond on-site endurance it characterizes (§5.1), with the evacuation that followed (§5.8). It asks whether the class's design basis anticipated what happened, and returns what it finds.

1.2 Position and Scope

A reference event instrument, in the series convention, reads a real event backward through the design basis, adds no requirements of its own, and returns findings and recommendations for disposition. This one is issued under the Healthcare and Public Health Sector Framework v0.6, inherits the DBA-MA parent framework, and was conferred by the framework's §9.1 on the author's decision of 12 October 2026, on the proposal of *Care Delivery*. On the author's decision of the same day it is an event distinct from *The Failure of Coordination in Hurricane Katrina* (DBA-ESS-RE-8): this is one hospital's loss of power and what followed for the patients inside, and the regional context, the levees, the flood and the failure of the response, is that instrument's and is cited, not restated (§5.1, §10.7). Section references of the bare form §N are to *Care Delivery* v0.2, the version this instrument read.

The scope runs from landfall on 29 August 2005 to the departure of the last living patient on the evening of 1 September, with the hospital's plans and the warnings that preceded the storm. Times are Central Daylight Time. The record of the event is closed.

The attribution posture is fixed. No individual is named: the hospital's physicians, nurses and staff, the patients and the people later arrested are named by role or not at all. A grand jury declined to indict on any count, and this instrument makes no finding on the deaths, on cause beyond what its sources state, or of fault. The hospital was owned by Tenet Healthcare; a separate company, LifeCare, ran a long-term acute care hospital on its seventh floor.

The hospital, its essential electrical system and the care given in it lie within the care-delivery class (§1.2). Its loss of grid power and water came through the energy and water sectors (§5.1), proposed as impacted sectors at Section 5.2. The Foundational Definitions are cited by designation, FD-REG holding their versions; none is applied here, for the reason given at Section 6.10.

2. Sources and Attribution

2.1 Sources and Their Limit

The primary source is ProPublica's investigation, "The Deadly Choices at Memorial" (Sheri Fink, ProPublica with the New York Times Magazine, 27 August 2009), read to most of its length, with ProPublica's report of the class action brought for those inside the hospital, which states the experts' positions on why the power failed. Neither is a board's or regulator's report; no such report on the hospital was found. *Care Delivery* §5.1 reads the event from the same source. No non-public material is used.

2.2 Where the Sources Differ

On why the power failed, the two experts in the class action differed: the plaintiffs' expert concluded that floodwater shorted out the electrical transfer switches and many distribution panels, and Tenet's expert attributed the failure to mechanical problems in the diesel generators, which stood well above the water, as ProPublica reported. The investigation itself states that the generators "sputtered and stopped" and that the main transfer switches stood "only a few feet above ground level", and does not link the two. This instrument states both positions as their authors' and resolves neither (Finding F-1).

3. The Event

3.1 The Hospital and Its Site

The hospital stood "three feet below sea level" in a low part of the city (ProPublica). Its main emergency-power transfer switches were located "only a few feet above ground level", as at many hospitals in the flood zone. After Hurricane Ivan in 2004, facilities staff warned that "it won't take much water in height to disable the majority of the medical center"; moving the electrical equipment would have been costly, and only cheaper improvements were made; a recommendation of 2004 to move parts of the system above the ground floor was not acted on (ProPublica). The hospital's emergency plan, 246 pages long, "offered no guidance for dealing with a complete power failure", and none for evacuating if the streets flooded; staff expected rescuers to evacuate everyone within a few hours. The plan assumed that backup power would last at least three days (ProPublica, class action report).

3.2 The People Inside

At landfall about 2,000 people were in the hospital, more than 200 of them patients and about 600 of them workers. LifeCare's 52 patients were on the seventh floor.

3.3 The Loss of Power

City power failed at 04:55 on Monday 29 August; the generators took over, and the air conditioning shut down by design. On Tuesday 30 August the floodwater rose around the hospital, and evacuation by boat and by helicopter from a long-unused helipad on an eight-story parking garage, with minimal lighting and no guard rail, began that afternoon. At about 02:00 on Wednesday 31 August the generators stopped; the batteries of the ventilators ran out within about half an hour. Backup power had lasted less than two days against the three the plan assumed. Without power or running water, the temperature inside rose above 100 degrees. One elevator still worked for a time.

3.4 The Evacuation

The patient census fell from 187 to about 130 by Tuesday night. Evacuation continued through Wednesday and Thursday, with Tenet sending privately hired helicopters and Coast Guard helicopters carrying patients out; the investigation records an occasion on which the Coast Guard offered to evacuate more patients and those in charge declined. The last living patient left the hospital grounds at about 21:00 on Thursday 1 September.

3.5 The Dead

Forty-five bodies were recovered from the hospital, more than from any comparable hospital in the city (ProPublica). The sources read give no breakdown between Memorial's and LifeCare's patients. The deaths of some patients were investigated; a grand jury declined to indict, and no finding is made here.

4. Technical Decomposition

The event decomposes into initiating elements (I-1 to I-4) and effect elements (E-1 to E-5), as Table 4-1 sets out. Composed, they are the class's loss of electric power beyond on-site endurance (§5.1), in which the essential electrical system was lost to the event that took the grid, followed by an evacuation that took three days (§5.8), in Chain C, with the regional Chain S and the response that is DBA-ESS-RE-8's.

Table 4-1. Event elements and their mapping. Section references are to *Care Delivery* v0.2.

No.	What happened	Class provision	Section
I-1	Hospital three feet below sea level; switches a few feet above ground; 2004 warning not acted on	Components protected from the event (§5.1)	3.1
I-2	Plan with no guidance for total power failure or flooded streets; three days' backup power assumed	Endurance stated (§5.1); evacuation planned (§5.8)	3.1
I-3	Grid lost at landfall; flood around the hospital the next day	Loss of power (§5.1); water (§5.2)	3.3
I-4	Generators stop at about 02:00 on 31 August, cause contested	Essential electrical system lost to the event (§5.1)	2.2, 3.3

E-1	Ventilator batteries out within about half an hour; heat above 100 degrees	Continuity window, life-sustaining band (§2.2, §3.3)	3.3
E-2	Evacuation by boat and from an unlit helipad, one elevator, over three days	Time to evacuate by band (§5.8)	3.3, 3.4
E-3	A tenant hospital with its own patients on the seventh floor	Scope of the hospital's plan (§1.2)	3.2
E-4	Forty-five bodies recovered	Chain C	3.5
E-5	Regional collapse of the response	Cited from DBA-ESS-RE-8	1.2

5. Tier Determination

5.1 The Four Characteristics

Intentionality, persistence, precision and denial are absent. The event was a hurricane and the failure of the city's levees, as DBA-ESS-RE-8 reads the region; no act of any party bore on the hospital.

5.2 Determination

The event does not meet the threshold for the military-action extension on any of the four characteristics. It is a non-adversarial loss of electric power beyond on-site endurance (§5.1) at a hospital in a natural-hazard exceedance (§5.8), in Chain C. That is the tier determination returned for the DBA Reference Event Register. The impacted sectors proposed are Energy, for the grid, and Water, for the municipal supply, through which the class records the loss came (§5.1).

6. Characterization Against the Design Basis

6.1 §5.1 The Class's Account of the Event

Held, with a gap. Every figure checked holds, and the transfer switches' height is ProPublica's, as the conferral note says. Two points are qualified and go to the class as Erratum 2 of Project 11's note of 13 October 2026 (Healthcare batch): the class's sentence links the generators' stopping to the switches' height, which the investigation does not and the experts dispute; and the building's elevation, which the conferral note found missing, is three feet below sea level on ProPublica's account.

6.2 §5.1 The Essential Electrical System Lost to the Event

Held, with a gap. The category is realized as written: the system that was to carry the load was lost to the event that took the grid. Whether the switches or the generators failed, the class's requirement reaches both. The event adds what the plan lacked when the system failed. Finding F-1.

6.3 §5.1 and §5.8 The Site and the Warning Not Acted On

Held, with a gap. The class requires the components located and protected so that the event does not take them; the site's height against the flood was known and a warning recorded. Finding F-2.

6.4 §5.8 Evacuation

Held, with a gap. The class requires the time to move every patient, band by band, by means that survive the event. The means here were an unlit helipad reached by one elevator, then by hand. Finding F-3.

6.5 §1.2 and §5.8 The Hospital Within the Hospital

Held, with a gap. A separate company's hospital, with its own patients, occupied the seventh floor. Finding F-4.

6.6 §2.2 and §3.3 The Continuity Window

Held. The window of the life-sustaining band, once the system failed, was the ventilators' batteries, about half an hour; the class's band and its measure read the event.

6.7 §5.2 Loss of Water

Held. Running water was lost with the power; the class's statement that a hospital can lose its habitability through a utility reads it.

6.8 §6.8 and §6.9 The Autonomous Care Continuity Condition

Held. The second declared residual, a loss of power outlasting endurance ending in an evacuation the hospital cannot avoid, is what the event realized, from a storm and not a campaign.

6.9 §6 The Military-Action Extension

Not exercised. No act of any party bore on the event.

6.10 §7.3 and §7.4 The AI-Governance Design Basis

Not exercised. No component within the scope of *Layer Decomposition and Determinism* §4.3 bore on the event.

6.11 Summary of Design Basis Performance

Of the ten readings, five Held, with a gap, the class's account among them; three Held; and two were Not exercised. Nothing was Missed. The class's §5.1 reads the event, including its point that the weak link may not be the generator. What the event adds is the plan for the failure of the

system itself, the warning recorded and not acted on, the evacuation's means, and the tenant inside the walls.

7. Findings

Section references are to *Care Delivery* v0.2 unless another instrument is named.

7.1 F-1 — The Plan Covers the Failure of the Essential Electrical System Itself

The plan assumed backup power would last at least three days, it lasted less than two, and the plan "offered no guidance for dealing with a complete power failure"; once the generators stopped, the ventilators' batteries were the window, about half an hour (Sections 3.1 and 3.3). Why the system failed is disputed between the switches and the generators (Section 2.2). The finding confirms §5.1's requirement that every component between the generators and the load be located and protected so that the event does not take them, which reaches both positions, and is that the class should state that the hospital also plans for the loss of its essential electrical system, whatever its cause: the battery endurance of each life-sustaining device and the means of continuing its function by hand, the order in which patients dependent on power are moved, and the trigger for that move set before the system fails, so that the continuity window of the life-sustaining band after the system's loss is stated and not discovered.

Affected: §3.3, §5.1, §5.8. Disposition: R-1.

7.2 F-2 — The Site's Height Against the Flood Is Stated, and a Known Warning Is a Departure

The hospital stood three feet below sea level, its main transfer switches a few feet above ground; staff warned after Hurricane Ivan that little water would disable most of the center, and a recommendation of 2004 to raise parts of the electrical system was not acted on for its cost (Section 3.1). The finding confirms §5.1's required function, and is that the class should state that the hospital records the elevation of every component of its essential electrical system, its fuel and its water and medical gas plant against the design-basis flood of its site, and that a warning or recommendation that a component is below that flood, once made, is carried in the design basis as a departure with the protective measures that stand in for it, until it is corrected, as the dam class treats a shortfall in release capacity (*Dam Structures and Water-Control Works* §5.1).

Affected: §5.1, §5.8, §6.10. Disposition: R-2.

7.3 F-3 — The Means of Evacuation Are Part of the Design Basis

The patients left by boat and from a long-unused helipad on a parking garage, with minimal lighting and no guard rail, reached by one working elevator, and once the power was gone, by hand; the plan expected everyone out in a few hours, and the last living patient left more than

three days after landfall (Sections 3.1, 3.3 and 3.4). The finding confirms §5.8's requirement that the hospital state the time to move every patient band by band by means that survive the event, and is that the class should state that those means are named and their components held to the same protection as the essential electrical system: the route from each ward to the point of departure, the elevators and lighting it needs and the branch that powers them, the point of departure itself, and the time per patient by band on that route without power.

Affected: §5.1, §5.8. Disposition: R-3.

7.4 F-4 — A Tenant's Patients Are Inside the Hospital's Plan

A separate company ran a long-term acute care hospital, with 52 patients, on the seventh floor (Section 3.2), and ProPublica records evacuation offers between the two being declined. The finding confirms §1.2's scope, the facilities in which patients are cared for, and is that the class should state that every patient in the building, whoever operates the unit that holds them, is within the hospital's continuity window and its evacuation plan, under one incident command, with the tenant's obligations stated in the lease or agreement that admits it.

Affected: §1.2, §5.8, §8.3. Disposition: R-4.

8. Recommendation Register

Priority is on the series' one-to-three scale, 1 the most consequential. R-1 to R-4 dispose to *Care Delivery*.

No.	Title	Affected	Recommendation	Pri.
R-1	Plan for the loss of the essential electrical system itself	§3.3, §5.1, §5.8	State each life-sustaining device's battery endurance and manual continuation, the order and trigger for moving power-dependent patients, and the life-sustaining window after the system's loss.	1
R-2	State the site's height against the flood; carry a known warning as a departure	§5.1, §5.8, §6.10	Record every critical component's elevation against the site's design-basis flood; carry an unremedied warning as a departure with its stand-in measures.	1
R-3	Name and protect the means of evacuation	§5.1, §5.8	Name route, elevators, lighting and their branch, and point of departure; protect them as the essential electrical system; state time per patient by band without power.	1
R-4	Bring a tenant's patients inside the plan	§1.2, §5.8, §8.3	Every patient in the building is within the continuity window and evacuation plan under one incident command, the tenant's obligations stated in its agreement.	2

9. Open Items

OI-1: the register row. Project 0 is asked to enter the tier determination of Section 5.2 and the impacted sectors it proposes, and to record the link to DBA-ESS-RE-8 that the conferral note asks for. OI-2: no board's or regulator's report on the hospital was found; if one is, Sections 2.2 and 3.3 should be read against it. OI-3: the class has not taken in this instrument and conforms at its next revision, with Erratum 2.

10. Standing of This Document

10.1 What This Instrument Establishes

It modifies neither class nor framework. It makes no finding on the deaths, on why the power failed, or of fault.

10.2 Relationship to the Regional Instrument

The Failure of Coordination in Hurricane Katrina (DBA-ESS-RE-8) holds the region: the levees, the flood and the response. This instrument holds one hospital, and the two are read together for the hospital's evacuation, whose rescuers were the region's.

10.3 Relationship to the Class's Use Case

DBA-HPH-UC-1 (*A Mass-Casualty Surge During a Grid and Communications Outage*) drives the class's governing scenario with the hospitals on generator power through a campaign. Its surge and its campaign are its own; the reading this event offers it is F-1's, the window after the essential electrical system is lost.

References

Fink, S., "The Deadly Choices at Memorial", ProPublica with the New York Times Magazine, 27 August 2009. ProPublica, "Trial to Open in Lawsuit Connected to Hospital Deaths After Katrina", undated as read. The DBA instruments named in the text, with the Foundational Definitions and FD-REG, and the DBA-MA Parent Framework.

Revision History

Version 0.1, 13 October 2026: issued in Project 11 under the Healthcare and Public Health Sector Framework v0.6, on the conferral of the framework's §9.1 of 12 October 2026, read against *Care Delivery* v0.2, citing DBA-ESS-RE-8 for the regional context on the author's decision of the same day. Written new under the Reference Event Compression Standard v0.3. Tier determination at Section 5.2; F-1 to F-4, R-1 to R-4. The class's account holds, with two qualifications going to the class as Erratum 2 of Project 11's note of 13 October 2026 (Healthcare batch).